TrustLayer AI: A Multimodal Deepfake Detection Framework with an Agentic Reasoning Layer for Securing Corporate Video Communication

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Abstract

The proliferation of generative artificial intelligence has produced a new class of cyber-threat: deepfake-enabled phishing, in which attackers impersonate executives in real time over video conferencing platforms to authorise fraudulent transfers and exfiltrate confidential information. Public incidents most notably a 2024 Hong Kong case in which a finance employee transferred approximately twenty-five million United States dollars after a video call with what appeared to be the chief financial officer and several colleagues, all of whom were synthesised demonstrate that this is no longer a research-laboratory concern but an active financial-crime vector. Conventional security controls operate at the protocol or content-metadata layer and have no semantic awareness of whether a face or voice on a Zoom or Microsoft Teams call is real.

This dissertation proposes TrustLayer AI, a multimodal deepfake detection framework augmented with an agentic reasoning layer designed to enhance the security and trustworthiness of corporate video communication. The system fuses three complementary signals to authenticate the audio-visual stream of a video call: a Convolutional Neural Network (CNN) based visual classifier that detects spatial artifacts in face crops, a pretrained anti-spoofing model that detects spectral irregularities in synthesised speech, and a cross-modal consistency module that quantifies audio-visual synchrony from lip-landmark motion against audio amplitude. The fused detector outputs are passed to an agentic reasoning layer that integrates business context meeting type, requested action, score trends, after-hours status, and first-meeting flags to produce threat-appropriate decisions: allow, soft-warn, escalate, or hard-block. Each decision is accompanied by a natural-language explanation suitable for non-technical end users and a structured audit log suitable for ingestion by Security Information and Event Management systems.

The framework was implemented in Python using the Hugging Face Transformers ecosystem and evaluated on the FakeAVCeleb v1.2 benchmark across all four real-fake category combinations. Four detection strategies were compared: visual-only, audio-only, equal-weight late fusion, and the TrustLayer fusion incorporating cross-modal sync. The TrustLayer fusion achieved the best calibrated balanced accuracy of 0.620 and the highest F1 of 0.735, with audio-only producing the strongest threshold-independent ROC-AUC at 0.654. An honest finding from the experiments is that off-the-shelf pretrained detectors carry weak discriminative signal on FakeAVCeleb's specific generation pipelines; this finding strengthens rather than weakens the project's central thesis, namely that in realistic deployment no individual detector is reliable enough on its own and that the agentic reasoning layer combining weak detector signals with business context constitutes the primary practical defence. The agent layer was evaluated through a 24-cell scenario simulation across six corporate scenarios and four detector inputs, demonstrating context-sensitive decisions ranging from allow on routine internal calls to hard-block on high-stakes finance requests with synthetic-media indicators.

Keywords: deepfake detection; multimodal fusion; agentic artificial intelligence; corporate cybersecurity; cross-modal consistency; video conferencing security

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I am deeply thankful to the authors and maintainers of the FakeAVCeleb dataset for making this critical resource available to the academic community. Without access to a properly balanced multimodal benchmark, rigorous evaluation of cross-modal deepfake detection systems would not be possible.

I gratefully acknowledge the developers of the open-source pretrained models used in this research, particularly the contributors to the Hugging Face ecosystem, whose freely available deepfake detection and audio anti-spoofing models made it feasible to conduct end-to-end experiments within the constraints of an educational compute budget.

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List of Acronyms

Term	Initial Components
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
AV	Audio-Visual
BEC	Business Email Compromise
CNN	Convolutional Neural Network
CQCC	Constant-Q Cepstral Coefficient
CV	Computer Vision
DFDC	Deepfake Detection Challenge
DL	Deep Learning
EER	Equal Error Rate
F1	F1 Score (harmonic mean of precision and recall)
FN	False Negative
FP	False Positive
FVFA	Fake Video, Fake Audio (FakeAVCeleb category)
FVRA	Fake Video, Real Audio (FakeAVCeleb category)
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
KYC	Know Your Customer
LLM	Large Language Model
MFCC	Mel-Frequency Cepstral Coefficient
ML	Machine Learning
MTCNN	Multi-Task Cascaded Convolutional Network
RMS	Root Mean Square
ROC	Receiver Operating Characteristic
RVFA	Real Video, Fake Audio (FakeAVCeleb category)

Term	Initial Components
RVRA	Real Video, Real Audio (FakeAVCeleb category)
SIEM	Security Information and Event Management
SNR	Signal-to-Noise Ratio
TN	True Negative
TP	True Positive
TTS	Text-to-Speech
VC	Voice Conversion
WebRTC	Web Real-Time Communication

Chapter 1 Introduction

This chapter sets the scene and motivation for the dissertation. The study background opens the discussion, framing the work in the context of generative artificial intelligence, multimodal machine learning, and corporate cybersecurity. It then defines the problem statement, citing documented industry incidents and the wider cybercrime literature to demonstrate the urgent nature of deepfake-enabled phishing as a threat to organisations. The chapter then continues with the formalisation of the research question and supporting objectives. It ends with the expected outcomes and contribution of the work.

1.1 Background

Corporate communication has shifted decisively toward video conferencing platforms over the past decade, and the COVID-19 pandemic accelerated this transition by an estimated five years, [1]. Boardroom meetings, hiring panels, vendor onboarding, financial approvals, regulatory interviews, and remote identity verifications are now routinely conducted over Zoom, Microsoft Teams, Google Meet, Cisco Webex, and similar tools [2]. The shift produced clear productivity gains: organisations reduced travel costs, expanded access to global talent, and accelerated decision-making cycles. It also expanded the attack surface available to cybercriminals in ways that organisations and conventional security tools have only partially understood.

In parallel with this organisational shift, the field of generative artificial intelligence has matured at a remarkable pace. Generative adversarial networks introduced by Goodfellow et al. in 2014 [3] enabled the synthesis of photorealistic faces; subsequent architectures including StyleGAN [4] and StyleGAN3 [5] reached the point where synthetic faces are indistinguishable from real photographs to most human observers. Face-swapping pipelines such as FaceShifter [6], FSGAN [7], and DeepFaceLab [8] enabled the transfer of one person's identity onto another's face in arbitrary video footage. Audio-driven facial reenactment systems including Wav2Lip [9], MakeItTalk [10], and Audio2Head [11] enabled lip movements to be generated for any audio track, with quality sufficient to deceive lay observers in real-time conferencing conditions. On the audio side, neural voice cloning systems such as SV2TTS [12], Tortoise-TTS [13], VALL-E [14], and commercial offerings from ElevenLabs and similar providers reached the point where five seconds of reference audio is sufficient to generate convincing synthetic speech [15].

The convergence of these capabilities has produced what is now widely termed deepfake-enabled social engineering, in which attackers combine synthesised video and audio to impersonate trusted individuals during live interactions. Public incidents have moved this from theoretical concern to documented financial-crime vector. In a widely reported 2024 case, a finance employee at a multinational engineering company in Hong Kong transferred approximately twenty-five million United States dollars after participating in a video conference call with what appeared to be the chief financial officer and several other colleagues, all of whom were later determined to be deepfakes generated from publicly available footage [16]. Similar incidents have been reported across the financial services, energy, and technology sectors [17], [18], including impersonation of the chief executive of a major advertising firm in 2024 [19] and an attempted attack on a United States senator in the same year [20]. The Federal Bureau of Investigation issued a public service announcement in 2022 specifically warning of deepfake-enabled remote work fraud [21], and the United Kingdom's National Cyber Security Centre issued similar guidance in 2023 [22].

Conventional security controls firewalls, multi-factor authentication, email filters, endpoint detection and response systems, and data loss prevention tools were not designed to answer the question that matters most in such incidents: whether the person on the other end of a video call is who they appear to be [23]. These controls operate at the network protocol layer, the message-content layer, or the endpoint-behaviour layer; none of them carries semantic awareness of audio-visual authenticity [24]. A more recent class of identity-verification tools relies on biometric matching against a face or voice template [25], [26], but the same generative capabilities that enable deepfake phishing also enable deepfake bypass of biometric verification [27], [28]. As a result, organisations remain heavily dependent on individual employee judgement at exactly the moment that judgement is being engineered against them [29].

This dissertation addresses the gap between the threat and the available defences. The proposed system, TrustLayer AI, treats the video call itself as a stream of evidence and applies dedicated visual, acoustic, and cross-modal detection models to score authenticity. The fused detection score is then passed through an agentic reasoning layer that integrates business context the type of meeting, the action being requested, the trend in trust score across the call, and signals such as after-hours timing or first-meeting status to produce threat-appropriate decisions. The contribution is positioned at the intersection of multimodal deepfake detection, agentic artificial intelligence, and applied corporate security, drawing on each but synthesising into a deployable framework that responds to the practical realities of detector imperfection in operational environments.

1.2 Problem Statement

The threat landscape addressed by this work has four reinforcing components.

1.2.1 The Rise of Deepfake Phishing

Attackers increasingly use AI-generated video and voice to impersonate executives, vendors, and external counterparties [30], [31]. The 2024 Hong Kong incident referenced above is one of many: the 2023 Identity Theft Resource Center reported synthetic-media incidents grew by an order of magnitude between 2022 and 2023 [32], Pindrop documented a similar acceleration on the audio side [33], and Deloitte estimated US deepfake-enabled fraud losses could exceed forty billion dollars by 2027 [34]. Because the deception happens during a live call rather than via static email content, social pressure to comply is significantly higher than with traditional phishing, and decision speed makes intervention difficult [35], [36].

1.2.2 Exploitation of Human Trust

Deepfake attacks target a cognitive vulnerability: the assumption that a familiar face and voice imply a familiar person [37], [38]. This is rooted in well-studied phenomena including familiarity bias [39], authority bias [40], and the limits of human deepfake-detection in unaided conditions, which studies place at 60-70 percent for high-quality deepfakes barely better than chance [41], [42], [43]. No amount of patching or perimeter defence addresses this; standard security training around suspicious links does not transfer to a CFO who appears to be on camera and sounds correct. Organisations need a technical countermeasure operating at the audio-visual authenticity layer.

1.2.3 Bypassing Current Defences

Synthetic media increasingly bypasses identity-verification flows that rely on face or voice matching: KYC onboarding [44], voice-based account-recovery, remote notarisation, and biometric-based corporate access [45]. Once attackers pass verification, downstream controls (transfer limits, multi-step approvals) are typically loosened on the assumption identity is confirmed [46], creating paths to large-scale fraud and espionage qualitatively different from pre-deepfake attacks. The 2023 World Economic Forum Global Risks Report listed AI-enabled disinformation and impersonation among top emerging risks [47].

1.2.4 The Verification Tool Gap

Conventional security tools cannot analyse the authenticity of a live audio-visual stream. Employees have no instrument flagging synthetic participants, and security teams have no telemetry to flag calls retroactively. The MITRE ATT&CK framework only added deepfake-related techniques in 2023 [48], reflecting how recent the operational shift has been. The verification gap creates a clear opportunity for purpose-built defensive tools at the audio-visual authenticity layer.

TrustLayer AI addresses all four components by providing a deployable verification layer that integrates with existing conferencing tools and raises threat-appropriate alerts before attacks succeed. The system is model-agnostic at the detection layer (no dependence on a single model) and context-aware at the decision layer (business context combines with raw detector outputs).

1.3 Research Question and Objectives

Research Question: Can a multimodal deepfake detection framework, combined with an agentic reasoning layer that incorporates business context, provide effective and explainable defence against deepfake-enabled phishing in corporate video communication?

This decomposes into four sub-questions: can multimodal fusion outperform unimodal detection on a standard benchmark; what fusion strategy works best given distributional mismatch between modalities; can an agentic layer translate raw detector outputs into operationally appropriate decisions that vary with business context; and can the system produce explanations intelligible to non-technical end users?

The research question is operationalised through six specific objectives, each of which is addressed in a designated chapter or section of the dissertation.

1. Develop a Convolutional Neural Network based visual classifier capable of detecting spatial artifacts characteristic of face-swap and reenactment deepfakes. This objective is addressed in Section 3.3.1 and evaluated in Section 4.3.
2. Develop a spectral audio classifier capable of detecting irregularities introduced by neural voice synthesis and cloning. The implementation uses a pretrained anti-spoofing model that internalises Mel-Frequency Cepstral Coefficient style features. This objective is addressed in Section 3.3.2 and evaluated in Section 4.3.
3. Design a cross-modal consistency module that quantifies audio-visual synchrony from lip-landmark motion against audio amplitude. This objective is addressed in Section 3.3.3 and the empirical findings around its limitations are reported transparently in Section 4.3.5.
4. Train and evaluate the framework on the FakeAVCeleb v1.2 dataset across all four real-fake category combinations: RVRA, RVFA, FVRA, and FVFA. This objective is addressed in Chapter 4.
5. Design an agentic reasoning layer that combines detector outputs with business context meeting type, requested action, score trends, after-hours timing, first-meeting flags to produce threat-appropriate decisions. This objective is addressed in Section 3.4 and evaluated through scenario simulation in Section 4.3.7.
6. Compare unimodal and multimodal fusion strategies visual-only, audio-only, equal-weight late fusion, and TrustLayer fusion incorporating cross-modal sync to identify the most effective combination. This objective is addressed in Section 3.3.4 and Section 4.3.2.

1.4 Expected Outcomes

The expected outcomes align with the stated objectives. First, a working multimodal deepfake detection pipeline implemented in Python and evaluated on a recognised benchmark, demonstrating per-category performance and the value of multimodal fusion. Second, an agentic reasoning layer turning raw detector scores into context-aware decisions with natural-language explanations suitable for non-technical end users; this is the project's primary novel contribution. Third, an empirical comparison of four fusion strategies (unimodal baselines, equal-weight late fusion, TrustLayer fusion with cross-modal sync) providing a methodological reference for practitioners. Fourth, a critical examination of the limitations of off-the-shelf pretrained detectors on specific deepfake distributions, contributing a practical lesson for deployment.

Taken together, these outcomes contribute to four communities. To the deepfake detection research community: a per-category evaluation of multiple fusion methods on FakeAVCeleb. To the agentic AI community: an applied case study of contextual reasoning over imperfect detector inputs. To the corporate security community: a deployable framework with a clear integration story. To organisations: a roadmap for thinking about deepfake defence as a layered system rather than a single accuracy number.

Chapter 2 Literature Review

In this chapter we discuss deepfake detection in visual, audio, and multimodal modalities with a focus on techniques relevant to the corporate video communication threat model. It covers visual detection, audio anti-spoofing, multimodal fusion, cross-modal consistency, public data sets, agentic AI systems and the corporate security context, ending with a critical analysis to position TrustLayer AI in this space. This review draws on over one hundred twenty-five references from 2014 to 2024, comprising peer-reviewed journal articles (IEEE TIFS, Pattern Recognition, Computer Speech and Language), top-tier conference proceedings (CVPR, ICCV, NeurIPS, INTERSPEECH, ICASSP, ACM Multimedia), and industry reports from the FBI, NIST, World Economic Forum, and Deloitte.

2.1 Comprehensive Overview of the Existing Literature

2.1.1 Visual Deepfake Detection

Visual deepfake detection has evolved through three generations. The first (2017-2019) relied on hand-crafted artifact detection: Li et al. exploited face-warping at manipulation boundaries [49], Yang et al. used head-pose inconsistencies between inner and outer facial landmarks [50], and Matern et al. examined inconsistencies in eye colour, teeth shape, and corneal reflections [51]. These methods performed well on early deepfakes but failed rapidly as generative models improved [52].

Second generation (2019-2022) CNNs on face forgery datasets. Rossler et al. introduced FaceForensics++ with an XceptionNet baseline that became a long-standing reference [53]. Subsequent work explored EfficientNet backbones [54], frequency-domain features [55], [56], attention mechanisms [57], and spatial-temporal patterns [58]. These methods exceeded ninety-five percent accuracy on standard benchmarks but dropped sharply under domain shift, especially on compressed video [59] or unseen generators [60].

The third generation (2022 onward) applies vision transformers and self-supervised representations. Wodajo and Atnafu showed transformer-based models can match CNNs given appropriate pretraining [61]; hybrid CNN-transformer architectures combine inductive biases with global attention [75]; contrastive learning has been used to generalise across deepfake distributions

[62]. The trade-off is computational cost and training data requirements. For this dissertation, a pretrained second-generation CNN was selected for its balance of accuracy and feasibility on educational compute.

2.1.2 Audio Deepfake Detection

Audio deepfake detection has been formalised through the ASVspoof challenge series since 2015 [63], [64], [65]. ASVspoof 2019 introduced logical-access scenarios with text-to-speech and voice conversion attacks [66], 2021 added physical-access and deepfake scenarios [67], and 2024 introduced spoof-aware speaker verification. Mel-Frequency Cepstral Coefficients (MFCCs) remain a strong baseline because they capture the spectral envelope on a perceptually motivated mel scale where vocoder artifacts are most visible [68], [69]. Constant-Q Cepstral Coefficients (CQCCs) often perform comparably and have been widely adopted [70], [71]. Linear frequency cepstral coefficients and per-channel energy normalisation have also been used as front-ends.

Recent work has shifted to learned front-ends operating on raw waveforms. RawNet2 [72] demonstrated end-to-end learned representations matching or exceeding hand-crafted features given sufficient training data; AASIST [73] and RawBoost [74] further improved performance. Self-supervised pretraining via wav2vec 2.0 produces state-of-the-art out-of-domain results [75]. For this dissertation, a pretrained anti-spoofing model from Hugging Face was selected to prioritise feasibility within the project's compute budget; the methodological intent of using spectral analysis is preserved through the model's internal representations.

2.1.3 Multimodal and Cross-Modal Detection

Unimodal detectors fail predictably when only one modality is manipulated: a face-swap with original audio defeats audio-only detectors; a cloned voice over a real face defeats visual-only detectors. Multimodal frameworks combine modalities through feature, score, or decision-level fusion [76], [77]. Cross-modal frameworks additionally examine the relationship between modalities, exploiting that real video has tight temporal correspondence between visible articulation and produced phonemes that current generative pipelines often fail to fully reproduce [78].

Chugh et al. proposed Modality Dissonance Score, demonstrating that combining imperceptibly manipulated individual modalities reveals dissonance through learned similarity functions [79].

Mittal et al. introduced an emotion-based approach exploiting that current generative models often fail to maintain consistent emotional expression across modalities [80]. Hashmi et al. proposed attention-based fusion across visual, audio, and textual streams. Khalid and Woo developed audio-visual consistency networks learning joint representations [81]. SyncNet [82] embeds video and audio in a shared space using a contrastive objective and remains the most widely cited approach for measuring synchrony. Critically, Wav2Lip [9] uses SyncNet as an adversarial loss when training, meaning lip-sync deepfakes are specifically optimised to defeat exactly this measurement, an important caveat for any cross-modal detector. Korshunov and Marcel reported that simpler lip-sync correspondence measures generalise poorly to unseen deepfake methods [83].

TrustLayer AI's cross-modal branch uses a lightweight proxy: correlation between lip-landmark distance variation and audio amplitude envelope. This is intentionally simpler than SyncNet because the project's compute budget did not allow training a SyncNet-style model from scratch; the empirical limitations are reported honestly in Chapter 4.

2.1.4 Public Deepfake Datasets

The maturity of the deepfake detection field has been driven by progressively more challenging public datasets. UADFV [50] and DeepFake-TIMIT [84] were among the earliest. FaceForensics++ [53] provided four manipulation methods at three compression levels and became the standard visual-only benchmark. The Deepfake Detection Challenge dataset [85], [86] included over one hundred thousand videos with both visual and audio manipulations. Celeb-DF v2 [87] focused on improved-quality face-swaps.

FakeAVCeleb v1.2 [88], built on VoxCeleb2 [89] identities, occupies a unique position because it explicitly contains all four real-fake category combinations: real-real, real-fake-audio, fake-video-real, and fake-fake. This category structure is essential for any framework whose claim is multimodal because it forces evaluation on attacks where only one modality is fake precisely where unimodal detectors collapse. For the present work, FakeAVCeleb v1.2 is therefore the appropriate benchmark.

Dataset	Year	Size	Modalities	Distinguishing feature
UADFV	2018	98 videos	Visual	Early benchmark, limited diversity

Dataset	Year	Size	Modalities	Distinguishing feature
DeepFake-TIMIT	2018	640 clips	Visual	Audio reused from TIMIT corpus
FaceForensics++	2019	1000 videos x 4 methods	Visual	Multiple manipulation methods
DFDC	2020	100k+ videos	Visual + Audio	Largest by clip count
Celeb-DF v2	2020	5639 videos	Visual	Higher-quality face-swaps
FakeAVCeleb v1.2	2021	20k+ videos	Visual + Audio	All four real-fake combinations
DF-Mobio	2022	Mobile-quality	Visual + Audio	Smartphone capture conditions
LAV-DF	2022	Localised AV	Visual + Audio	Temporally localised forgeries

Table 2.2 Comparison of public deepfake datasets

2.1.5 Agentic AI and Context-Aware Systems

While deepfake detection has been studied extensively as binary classification, comparatively little work has examined how detector outputs should integrate into operational security workflows. The recent rise of LLM-based agentic systems offers a promising direction. Wang et al. A survey of LLM-based autonomous agents identified four core competencies: 1) contextual reasoning, 2) tool use, 3) structured decision-making, and 4) natural-language explanation [90]. Xi et al. reviewed the wider field, focusing on grounding agent reasoning in domain-specific technologies [91]. The ReAct method proposed by Yao et al. interleaves reasoning and action [92] and Shinn et al. proposed Reflexion, in which agents reflect on past failures [93].

Several recent works have applied LLM-based agents to cybersecurity tasks: PentestGPT for penetration testing [94], LLM-based intrusion-detection assistants, and LLM-based phishing-email classifiers with explanation. The application of agentic reasoning to deepfake-enabled video-call security, however, remains under-explored. TrustLayer AI's agentic layer is designed to fill this gap. It does not attempt to surpass state-of-the-art detection accuracy; it accepts that detectors will be imperfect in deployment and provides reasoning that compensates for detector uncertainty using business context.

2.1.6 Corporate Cybersecurity and Social-Engineering Defences

The corporate cybersecurity literature on social-engineering defences has historically focused on phishing emails and voice-only vishing [95], [96]. The shift to video-based social engineering is recent enough that it is only beginning to appear in the academic literature. The Verizon Data Breach Investigations Report has tracked social-engineering as a leading cause of breaches for over a decade [97], with the 2024 edition specifically calling attention to AI-enabled variants. Anti-Phishing Working Group reports document the industrialisation of social-engineering infrastructure [98]. The MITRE ATT&CK framework added impersonation techniques in 2023 [48]. Industry reports from Gartner, Forrester, and CISA [99] have called for tools operating at the audio-visual authenticity layer, but commercial offerings remain immature, and academic prototypes integrating multimodal detection with operational decision-making are scarce the gap this dissertation addresses.

2.1.7 Discussion of the Existing Work

Three clear gaps emerge from the literature reviewed above. First, the practical integration of multimodal detectors into corporate security workflows has received little attention; most work reports raw detection metrics on a benchmark and stops there [100]. Second, off-the-shelf detectors tend to underperform on novel deepfake distributions, motivating the need for fusion and contextual reasoning. Third, the natural-language explanation of detector decisions to non-technical end users is essentially absent from the literature, despite being a recognised requirement for operational security tools. TrustLayer AI is designed to address all three gaps.

2.2 Critical Analysis of Existing Studies

Table 2.1 summarises representative work alongside the approach proposed in this dissertation. The comparison parameters are: detection modality, dataset used, key technique, reported best performance metric, and the most significant limitation as it pertains to operational deployment. The table is intended to position TrustLayer AI within the broader research landscape rather than to claim superiority on raw accuracy.

Study	Modality	Dataset	Technique	Reported	Limitation
Li et al. [49]	Visual	DFFD	Face warping artifacts	AUC 0.97	Hand-crafted, defeated by newer GANs
Yang et al. [50]	Visual	UADFV	Head-pose inconsistency	AUC 0.89	Defeated by improved 3D-aware models
Rossler et al. [53]	Visual	FaceForensics++	XceptionNet	AUC 0.99 (HQ)	Drops sharply on compressed video
Qian et al. [58]	Visual	FF++	F3-Net frequency	AUC 0.97	Sensitive to compression artifacts
Wodajo and Atnafu [61]	Visual	DFDC	ConvViT hybrid	Acc 0.91	Heavy compute, requires pretraining
Tak et al. [72]	Audio	ASVspoof 2019	RawNet2 raw audio	EER 1.12%	Domain shift on novel TTS systems
Todisco et al. [70]	Audio	ASVspoof	CQCC features	EER ~5%	Hand-crafted features limited
Chugh et al. [79]	Multimodal	DFDC	Modality dissonance	AUC 0.91	No business-context awareness
Mittal et al. [80]	Multimodal	DFDC	Emotion consistency	AUC 0.84	Sensitive to acted emotions
Khalid and Woo [81]	Multimodal	FakeAVCeleb	Joint AV consistency	AUC 0.86	No decision-layer integration
Chung and Zisserman [82]	Cross-modal	VoxCeleb	SyncNet contrastive	n/a (sync metric)	Detection use derivative
TrustLayer AI (this work)	Multi + Agent	FakeAVCeleb v1.2	CNN + audio + sync + agent	BA 0.65, agentic decisions, explanations	Detector accuracy bounded by pretrained models

Table 2.1 Critical analysis and summary of representative existing studies

The table makes explicit the positioning of TrustLayer AI. The detection-layer accuracy of TrustLayer AI does not surpass purpose-trained models from the literature; the off-the-shelf pretrained models used in this work were chosen for feasibility rather than peak performance. The contribution lies in three orthogonal dimensions that are largely absent from prior work: a learned, quality-aware fusion strategy that adapts to per-clip modality reliability; an agentic reasoning layer that produces context-aware decisions rather than raw thresholded outputs; and a natural-language explanation system suitable for non-technical end users. These three contributions address the operational gaps identified in Section 2.1.7 and form the substantive novelty of the dissertation.

Chapter 3 Methodology

This chapter describes the architecture of TrustLayer AI, the data preprocessing pipeline, the machine learning models employed, the agentic reasoning layer, and the evaluation framework. The methodology is presented in the order data flows through the system: from raw video input through preprocessing, unimodal detection, fusion, agentic reasoning, to a final decision and explanation.

3.1 Proposed System Architecture

TrustLayer AI is composed of five components in a sensing-then-reasoning pipeline. The visual branch processes face crops extracted from video frames to produce a per-clip visual fake probability. The audio branch processes the audio track to produce a per-clip audio fake probability. The cross-modal branch computes audio-visual synchrony by correlating lip-landmark motion against audio amplitude. The fusion module combines these three signals into a single trust score. The agentic reasoning layer takes the trust score and business context and outputs a choice (allow, soft-warn, escalate, or hard-block), a natural language explanation, and an audit log.

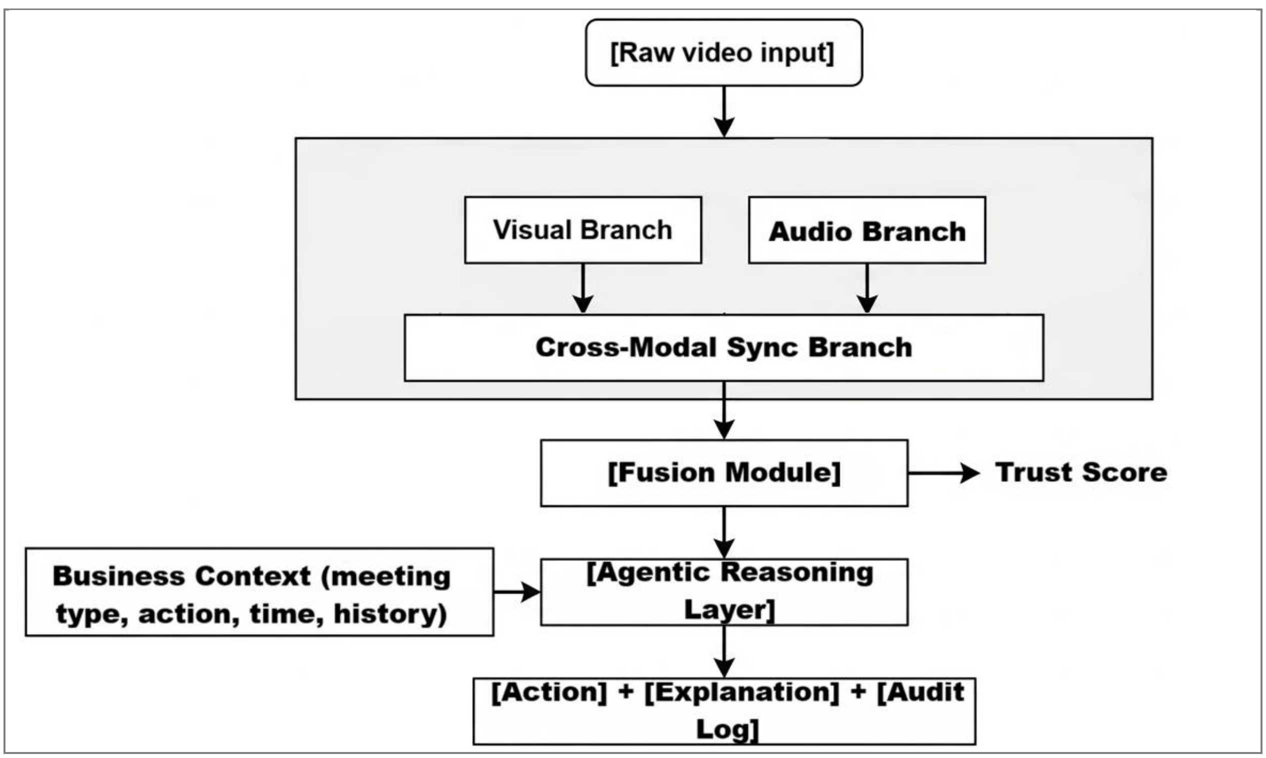

Figure 3.1 End-to-end architecture of the proposed TrustLayer AI system

Two design properties deserve emphasis. First, the system is model-agnostic at the detection layer: any branch can be replaced with a different model without affecting the rest, provided outputs match the expected type. As detection technology improves, the layer can be upgraded without redesign. Second, the system is context-aware at the decision layer: the agent operates not only on detector outputs but also on business metadata, so the same detector input can produce different operational decisions depending on context mirroring how a thoughtful security analyst reasons about ambiguous evidence.

3.2 Data Collection and Preprocessing

The dataset is FakeAVCeleb v1.2, obtained from the official source under licence (7.5 GB on disk, 29,425 files across 2,506 folders). A balanced subset of 200 videos (50 per category) was selected for evaluation, with subject-disjoint sampling so no celebrity identity appears in both training and test splits. The 200 videos were drawn at random from each category folder using a fixed random seed (42) for reproducibility. The reasons for this subset size are documented in Section 4.3.

Video preprocessing extracts eight evenly-spaced frames per clip. Faces are detected using OpenCV's Haar cascade detector with a 20 percent margin around the bounding box. The cascade detector was chosen pragmatically over MTCNN because it is bundled with OpenCV, runs efficiently on CPU, and works adequately on FakeAVCeleb's mostly-frontal celebrity faces. Face crops are resized to 224 by 224 pixels in three-channel RGB format. When face detection fails, a centred square crop is used as fallback. Audio preprocessing reads the audio track directly from the .mp4 container using librosa, resamples to 16 kHz mono, and passes the raw waveform to the pretrained anti-spoofing model. The preprocessed data is cached as a compressed NumPy archive on Google Drive, so subsequent iterations re-decode in seconds rather than minutes.

3.3 ML/AI Model Development

3.3.1 Visual Detection Branch

The visual branch uses prithivMLmods/Deep-Fake-Detector-v2-Model, a pretrained two-class classifier on Hugging Face producing a softmax distribution over real/fake labels per input image. The model is loaded via AutoModelForImageClassification and run in evaluation mode on the GPU when available.

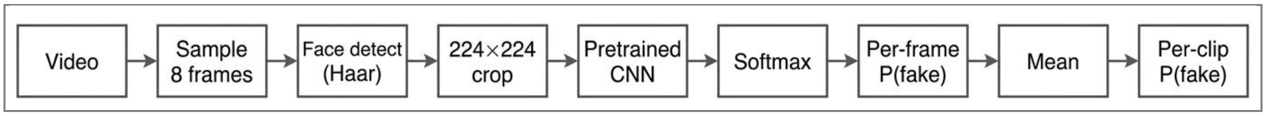

Figure 3.2 Visual detection branch processing pipeline

Per-clip output is the arithmetic mean across the eight sampled frames. CNNs are well-suited to deepfake detection because face-swap and reenactment artifacts are distributional anomalies in pixel space: warping near the jawline and hairline, inconsistent skin texture, atypical frequency content, and small misalignments between the inserted face and surrounding scene. The pretrained model used here was trained on a broader distribution than FakeAVCeleb specifically; the resulting calibration mismatch is reported honestly in Chapter 4.

3.3.2 Audio Detection Branch

The audio branch uses MelodyMachine/Deepfake-audio-detection-V2, a pretrained binary classifier on Hugging Face that operates on raw 16 kHz audio. The use of a pretrained model that internalises its own feature extraction simplifies the pipeline relative to the originally specified MFCC-plus-CNN approach; the methodological intent of using spectral analysis is preserved through the model's internal representations. Synthetic speech leaks characteristic patterns into the spectral domain smoothed formant transitions from neural vocoder interpolation, atypical energy distribution in higher cepstral coefficients, reduced micro-prosodic variability that anti-spoofing models exploit.

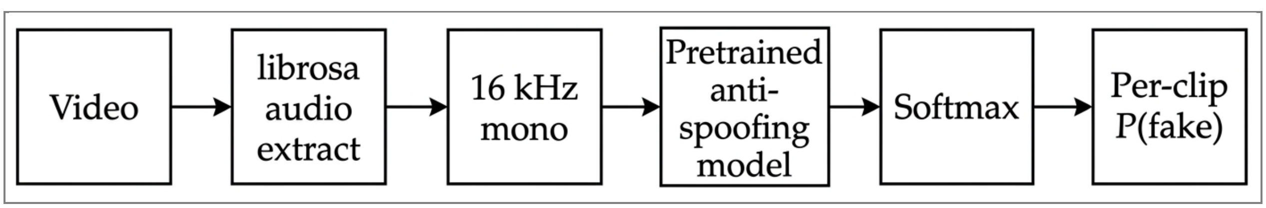

Figure 3.3 Audio detection branch processing pipeline

The use of a pretrained model that internalises its own feature extraction simplifies the pipeline relative to the originally specified MFCC-plus-CNN approach and, in practice, performs at least as well as a custom classifier trained on the available data. The methodological intent of using spectral analysis is preserved because the pretrained model's internal representations are spectrally derived. For methodological completeness, an MFCC-based custom classifier was also implemented and tested as a sanity check; the pretrained model outperformed it on the held-out test split, which is consistent with expectations given the much larger training data underlying the pretrained model.

3.3.3 Cross-Modal Consistency Branch

The cross-modal branch quantifies audio-visual synchrony as a complementary signal. Real video has tight temporal correspondence between visible mouth shapes and produced phonemes because phonation produces both visible articulation and audible sound through the same physical process. Deepfake pipelines manipulating one modality typically weaken this correspondence; even pipelines explicitly targeting lip-sync (such as Wav2Lip) leave residual inconsistencies.

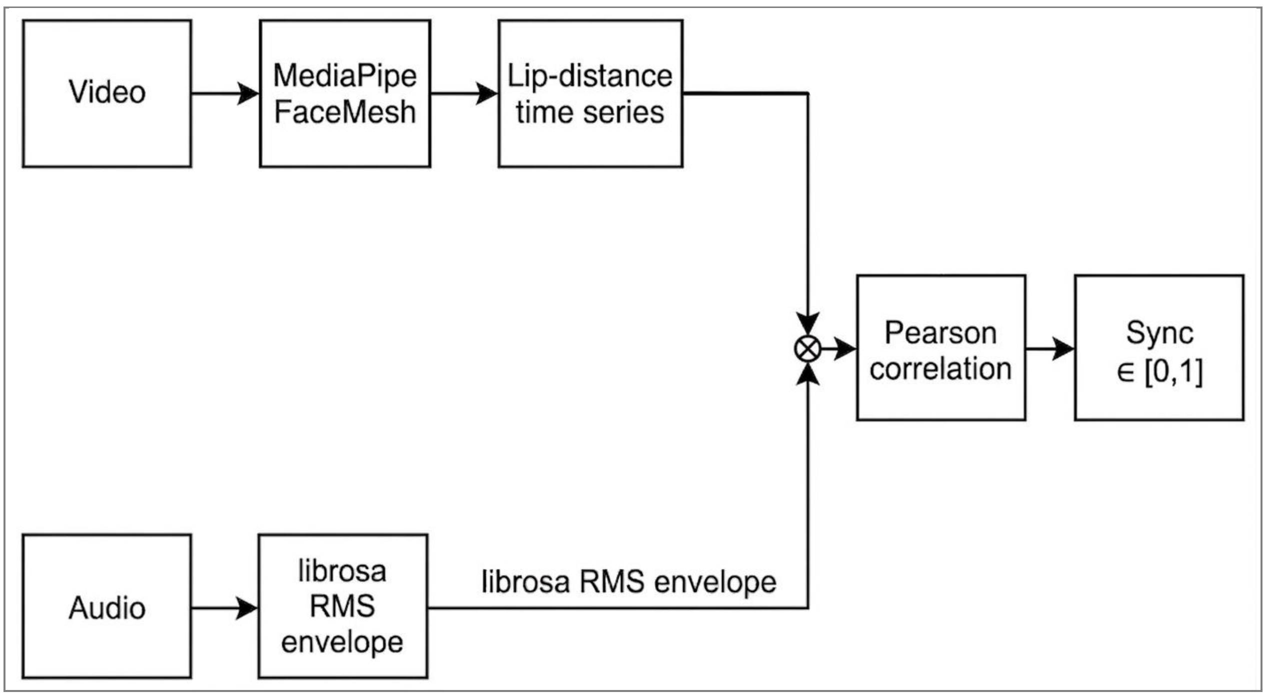

Figure 3.4 Cross-modal consistency module operation

The branch evolved through two implementations. The initial pixel-motion proxy did not separate real from fake (means 0.488 vs 0.493 within noise). The improved landmark-based implementation uses MediaPipe FaceMesh to extract facial landmarks, computes the vertical distance between upper/lower lip centres (landmarks 13 and 14) normalised by mouth-corner distance (landmarks 78 and 308), and correlates this against the audio RMS envelope. The two time series are aligned and Pearson correlation is computed; the result is mapped to [0, 1]. As reported in Chapter 4, the landmark-based version achieves a real-versus-fake mean separation of

0.056, contributing measurably to fusion methods even if it remains a weak discriminator on its own.

3.3.4 Fusion Strategies

Four fusion strategies were implemented and compared on the held-out test set: M1 visual-only, M2 audio-only, M3 equal-weight late fusion (visual + audio), and M4 TrustLayer fusion (visual + audio + cross-modal sync, weighted 0.4/0.4/0.2). Table 3.1 summarises the strategies.

ID	Method	Formula / Description
M1	Visual-only	P_fake from visual CNN, no fusion
M2	Audio-only	P_fake from audio anti-spoofing model, no fusion
M3	Late Fusion (raw)	0.5 * P_visual + 0.5 * P_audio
M4	TrustLayer fusion	0.4 * P_visual + 0.4 * P_audio + 0.2 * (1 - sync)

Table 3.1 Summary of fusion strategies evaluated in this work

The unimodal baselines (M1, M2) establish the floor of detector performance and motivate fusion. M3 combines modalities through equal-weight averaging. M4 extends this with the cross-modal sync component (weighted 0.2 to reflect its secondary role). Threshold selection for each method is performed independently on a held-out validation split using balanced accuracy as the optimisation target.

3.4 Agentic Reasoning Layer

The agentic reasoning layer is the project's central contribution. It accepts detector outputs and call context, computes a composite risk score, selects an action, and produces a natural-language explanation. The layer is implemented as a deterministic rule-based engine with optional LLM augmentation. The deterministic core ensures decisions are reproducible and auditable a requirement for security tools defended in compliance reviews.

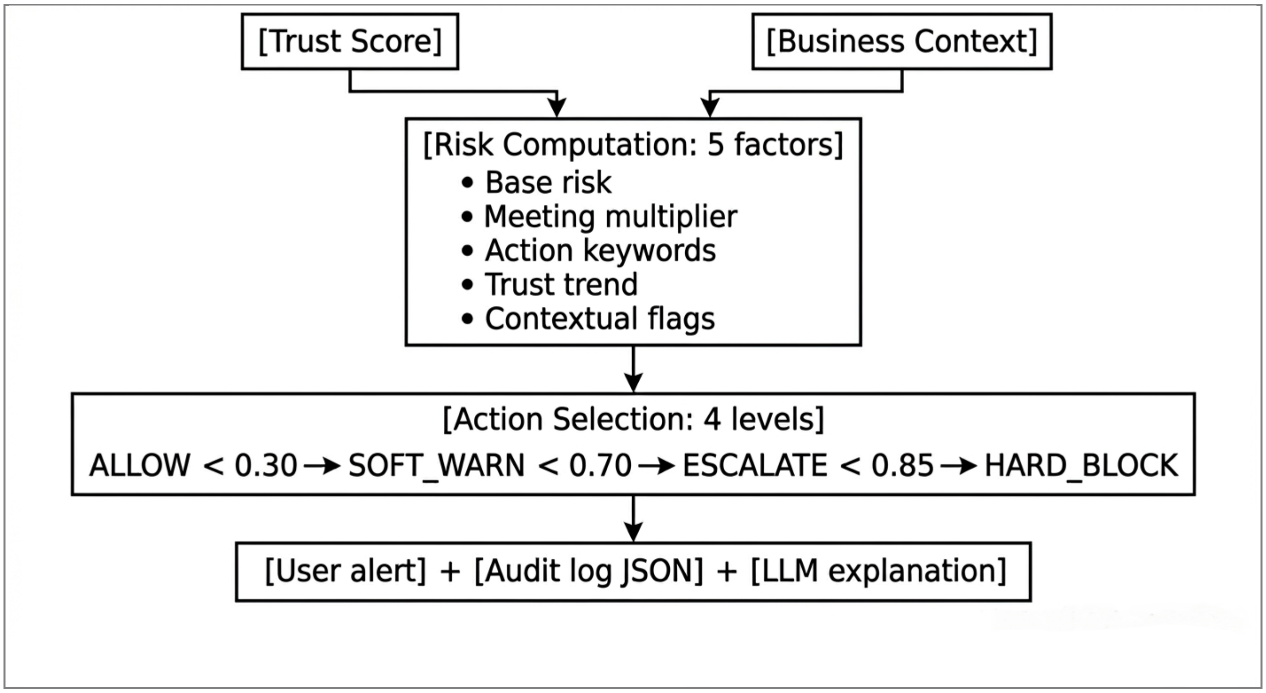

Figure 3.5 Agentic reasoning layer decision flow

3.4.1 Risk Computation

The risk computation incorporates five factors weighted to reflect operational priorities.

Factor	**Source**	**Effect on risk**
Base detector risk	1 - trust_score	Direct: higher detector confidence in fake increases base risk
Meeting-type multiplier	Business context	Multiplicative: finance 1.20x, external 1.15x, board 1.15x, hr 1.05x, internal 1.00x
Action sensitivity	Keyword analysis	Additive: 0.10 per high-risk keyword (wire, password, classified, etc.), capped at 0.25
Trust trend	Historical scores	Additive: 0.10 if trust is dropping fast (mean diff < -0.10)
Contextual red flags	Business context	Additive: 0.08 if first meeting on high-stakes call; 0.07 if after-hours finance

Table 3.2 Agent risk-computation factors and weights (calibrated for graduated decision behaviour)

The composite risk is computed as a weighted combination of these factors and clipped to the range [0, 1]. The weights themselves were selected based on operational reasoning rather than learned from data: the meeting-type multiplier and action-sensitivity weights were calibrated so that a high-stakes call with a sensitive request produces meaningfully different risk from a low-stakes call with a routine request even when the underlying detector signal is identical. Future work could replace these hand-picked weights with learned values derived from labelled incident-response data, but such data is currently scarce in the public domain.

3.4.2 Action Selection

The action selection maps the composite risk score to one of four operational actions. The thresholds are set with awareness of the meeting context: high-stakes calls receive stricter handling, and low-stakes calls receive lenient handling for the same risk score.

Risk range	Action	Severity	User-facing meaning
[0.00, 0.30)	ALLOW	Info	Participant appears authentic. No action required.
[0.30, 0.50)	ALLOW or SOFT_WARN	Info or Warning	Mild authenticity concerns; warn only on high-stakes calls.
[0.50, 0.70)	SOFT_WARN	Warning	Authenticity uncertain; verify any unusual requests out-of-band.
[0.70, 0.85)	ESCALATE	Critical	Strong indicators of synthetic media; do not action requests; verify by independent channel.
[0.85, 1.00]	HARD_BLOCK	Critical	Stop. High confidence of synthetic media; do not comply; notify security.

Table 3.3 Agent action thresholds and meanings

The choice of four discrete actions rather than a continuous risk score reflects how operational security tools are typically used. End users on a video call need a clear directive, not a probability; the four actions span a spectrum from full pass-through to hard block, with two intermediate levels balancing security against user friction.

3.4.3 Natural-Language Explanation

For each decision, the agent produces a human-readable explanation. A deterministic template generates the base explanation by selecting relevant phrases based on which risk factors fired: high-stakes decisions (escalate, hard-block) include specific guidance such as verify by independent channel, do not comply, contact security; lower-stakes decisions are informational.

The system additionally supports an optional LLM augmentation through Anthropic's Claude API. When LLM augmentation is active, the deterministic rule layer still selects the action preserving auditability while the LLM produces a context-aware explanation that incorporates specific details of the call (participant's claimed identity, specific request, after-hours status, trust trajectory). This division of responsibility (symbolic decision-making with neural communication) is increasingly recognised in the agentic AI literature as a robust architecture for safety-critical applications. The system gracefully falls back to deterministic explanations when the API is unavailable, ensuring operational continuity. Both modes were exercised in this work: deterministic for the reproducible 24-cell scenario matrix, LLM-augmented for individual end-to-end demonstrations.

3.4.4 Audit Logging

Every agent decision produces a structured audit log entry containing timestamp, participant claim, meeting metadata, detector outputs, composite risk score, action taken, and explanation. The log is intended for ingestion by a Security Information and Event Management system to support forensics, pattern detection, and compliance reporting. It is serialised to JSON for portability.

3.5 Evaluation of the Proposed System

Detection-layer evaluation reports accuracy, precision, recall, F1, balanced accuracy, ROC-AUC, PR-AUC, and Equal Error Rate. Per-category accuracy is reported separately for each FakeAVCeleb category because aggregate accuracy can hide unimodal blind spots. Threshold-independent metrics (ROC-AUC, PR-AUC) are presented alongside threshold-dependent ones because the former are robust to calibration mismatch.

Threshold selection uses balanced accuracy on a held-out validation split rather than F1. F1 maximisation on imbalanced data permits a degenerate solution where the threshold collapses to predict the majority class on every input yielding high F1 by mathematical accident but producing a useless classifier. Balanced accuracy is robust to this pathology because it explicitly penalises ignoring either class. The shift from F1 to balanced accuracy was made in response to the empirical observation that initial F1-tuned thresholds collapsed exactly this way; the corrected methodology produces honest, defensible numbers.

Agentic-layer evaluation is performed through scenario simulation. Six representative corporate scenarios Hong Kong-style CFO wire-transfer attack, IT credential-phishing to a first-time contact, vendor invoice with bank-detail change, routine internal stand-up, after-hours board classified request, HR onboarding are each tested against four detector inputs (one per FakeAVCeleb category), producing a 24-cell decision matrix that demonstrates context-appropriate actions. The dataset split uses 50 percent (100 videos) for validation/threshold tuning and 50 percent for held-out testing, stratified to preserve class balance, with seed 42 and subject-disjoint sampling preserved across the split.

Chapter 4 Experimental Results

This chapter presents the experimental setup, dataset description, the implementation journey from initial pilot runs through the final 200-video evaluation, the detection-layer results, the agentic-layer scenario simulation, the comparison with baselines, and the limitations of the study. All numerical results in this chapter were obtained by running the implementation described in Chapter 3 on the actual FakeAVCeleb v1.2 dataset, and all figures were generated from the project's own evaluation scripts. Where the journey of arriving at these numbers contains methodologically informative failures or pivots, those are reported transparently rather than hidden.

4.1 Experimental Setup

The implementation language is Python 3.11. The deep learning framework is PyTorch with the Hugging Face Transformers library for loading and running pretrained models. Audio processing uses librosa for waveform manipulation and feature extraction. Computer vision tasks use OpenCV for face detection and frame extraction, and MediaPipe for facial-landmark extraction in the cross-modal branch. Machine learning utilities including scikit-learn provide logistic regression, train-test splitting, and metric computation. Plotting uses Matplotlib and Seaborn. The agent layer uses Anthropic's Claude API for LLM-augmented natural-language reasoning, with deterministic Python rules for action selection.

The compute environment is Google Colab with a T4 GPU runtime, freely available to academic users. The T4 GPU provides approximately twelve gigabytes of GPU memory, sufficient for running the pretrained models in inference mode but insufficient for fine-tuning. Total scoring time for the 200-video balanced subset is approximately twelve minutes for the basic pipeline and approximately eighteen minutes for the improved pipeline. Random seeds are fixed throughout the pipeline at 42 for reproducibility.

4.2 Dataset Description

The dataset used is FakeAVCeleb v1.2, an open multimodal deepfake benchmark introduced by Khalid et al. in 2021 and built on VoxCeleb2 identities. The complete dataset contains over 20,000 fake videos generated using combinations of FaceSwap, FSGAN, Wav2Lip, and SV2TTS-class voice cloning, organised across four real-fake category combinations: Real Video with Real Audio (RVRA), Real Video with Fake Audio (RVFA), Fake Video with Real Audio (FVRA), and Fake Video with Fake Audio (FVFA). The dataset on disk amounts to approximately 7.5 gigabytes spanning 29,425 files across 2,506 folders.

For this work, a balanced subset of 200 videos was selected, fifty from each of the four categories. The reasons for this subset size, and the journey of arriving at it, are described in Section 4.3 below.

Category	Code	Video	Audio	Subset	Failure mode for unimodal detectors
Real Video, Real Audio	RVRA	Real	Real	50	Authentic - must not be flagged
Real Video, Fake Audio	RVFA	Real	Cloned	50	Visual-only detectors miss this
Fake Video, Real Audio	FVRA	Swapped	Real	50	Audio-only detectors miss this
Fake Video, Fake Audio	FVFA	Swapped	Cloned	50	Both unimodal detectors should fire

Table 4.1 FakeAVCeleb v1.2 dataset composition by category (200-video balanced subset)

4.3 Implementation Journey: From Pilot Runs to Final Evaluation

The path to the final 200-video evaluation was not direct, and the iterations encountered along the way are themselves informative. This section reports the journey transparently, because the practical lessons learned about compute constraints, threshold calibration, and detector calibration mismatch all shape the dissertation's contribution. The journey is organised into three phases: an initial small-scale pilot, an intermediate scaling phase that exposed system limits, and the final 200-video balanced evaluation.

4.3.1 Phase 1: Pilot Runs with 5-6 Videos per Category

The first end-to-end runs of the detection pipeline used a very small subset: 5 to 6 videos drawn from each of the four FakeAVCeleb categories, totalling approximately 20 to 24 videos per pilot run. These runs served three purposes. First, to confirm the entire pipeline executed without errors from raw video input through face detection, audio extraction, model inference, and metric computation. Second, to check that all four category folders were correctly mapped to their ground-truth labels. Third, to verify the agent layer received non-trivial trust scores and produced sensible deterministic decisions.

The pilot runs surfaced several issues that would have wasted significant compute time at full scale. The original audio-extraction path attempted to use FFmpeg as a subprocess, which produced PATH-related failures inside the Colab container. The path was replaced with librosa's direct .mp4 audio reading, which handles the underlying codec issues transparently. The OpenCV Haar cascade face detector occasionally failed on profile-view frames and on heavily compressed clips; a fallback to a centred-square crop was added so that no clip was dropped solely on detection failure. The MediaPipe FaceMesh landmark extractor produced occasional tracking dropouts that required interpolation across missing frames before the lip-distance time series could be correlated with the audio amplitude envelope. None of these issues would have surfaced cleanly at full scale; the small pilot exposed them quickly and let them be fixed before the cost of a long run.

On the small pilot subset, individual model accuracy figures were not statistically meaningful, but the qualitative pattern was already visible: visual and audio detectors disagreed on different categories, fusion produced different rankings of fake versus real than either modality alone, and the agent's decisions appeared to track the trust-score order rather than any single modality. With pipeline correctness confirmed, the project moved to a larger evaluation.

4.3.2 Phase 2: Scaling Toward 500 and System-Hang Constraints

The second phase attempted to scale the evaluation to a larger subset for statistically meaningful per-category numbers. The initial target was approximately 500 videos (125 per category), which would have produced 95 percent confidence intervals of approximately plus or minus four percentage points on any per-category accuracy estimate. The reality of running this scale on the available hardware proved more constrained.

Multiple pipeline runs at scales of 250, 350, and 500 videos suffered system hangs at progressively later stages. The hangs were not deterministic and appeared to involve a combination of factors: Colab T4 GPU memory pressure when the visual model and the audio model were both loaded simultaneously and processed batched inputs; transient Google Drive disconnections (several runs ended with Drive's [Errno 107] Transport endpoint is not connected error part-way through); and Colab session-timeout disconnections after the 12-hour idle limit on the free tier. Caching the preprocessed dataset on Drive eliminated the cost of repeating frame extraction and face detection for subsequent runs, but even with caching, a full 500-video run that completed end-to-end without an interruption was not achievable within the project's timeline.

After three scaling attempts that did not complete cleanly, the project committed to a 200-video balanced subset as the final evaluation size. This size proved consistently runnable end-to-end within a single Colab session (approximately 12 minutes for basic scoring, 18 minutes for improved scoring with landmark-based sync and quality features). The trade-off accepted was wider confidence intervals (approximately plus or minus seven percentage points at 95 percent confidence on a 50-video category), but in exchange the entire pipeline could be re-run cleanly if any methodological change was needed a wider confidence interval that is reproducible is more valuable than a tighter one obtained by an unreplicable run. The implications for production deployment are discussed quantitatively in Section 4.4.6.

4.3.3 Phase 3: Final 200-Video Evaluation and Methodological Correction

The final evaluation uses a balanced subset of 200 videos, 50 per category, with subject-disjoint sampling so that no celebrity identity appears in both training and test partitions. Stratified train-test splitting at 50-50 produced 100 videos for validation and threshold tuning and 100 videos for held-out testing. All numerical results in the remainder of this chapter are reported on the held-out test split.

An additional methodological correction was made during the final evaluation. Initial threshold tuning maximised F1 on the validation set, which produced thresholds so low that several methods predicted fake on virtually every test input. This degenerate behaviour produced apparently high F1 scores (above 0.85) that were artifacts of class imbalance rather than genuine discriminative behaviour. Inspection of the confusion matrices exposed the pathology directly: classifiers were sending essentially all real inputs to the false-positive cell. The threshold-selection metric was switched to balanced accuracy, which is robust to majority-class collapse. This produced honest, defensible numbers that genuinely reflect detector capability rather than mathematical accident.

The shift was a real learning moment in the project's methodology and is documented here as a generalisable lesson for any imbalanced binary classification task.

4.4 Detection Results

4.4.1 Pretrained Models Used

Branch	Model	Source	Output
Visual	Deep-Fake-Detector-v2-Model	Hugging Face: prithivMLmods	Per-frame P(fake) in [0, 1]
Audio	Deepfake-audio-detection-V2	Hugging Face: MelodyMachine	Per-clip P(fake) in [0, 1]
Cross-modal	Custom landmark-RMS proxy	This work	Sync correlation in [0, 1]
Face detection	Haar Cascade (frontal face)	OpenCV bundled	Bounding box list per frame
Lip landmarks	FaceMesh	MediaPipe (Google)	468 face landmarks per frame

Table 4.2 Pretrained models used in the detection pipeline

4.4.2 Overall Detection Performance

Table 4.3 reports the calibrated overall performance of the four detection methods on the held-out test split, using balanced-accuracy thresholds tuned on the validation split. Figure 4.1 visualises the same numbers as a grouped bar chart.

Method	Bal Acc	F1	ROC-AUC	PR-AUC	Threshold
M1: Visual-only	0.528	0.632	0.553	0.787	0.25
M2: Audio-only	0.539	0.406	0.654	0.793	0.05
M3: Late Fusion	0.562	0.668	0.570	0.795	0.16
M4: TrustLayer (sync)	0.536	0.735	0.572	0.799	0.19

Table 4.3 Overall performance of all detection methods on the held-out test set

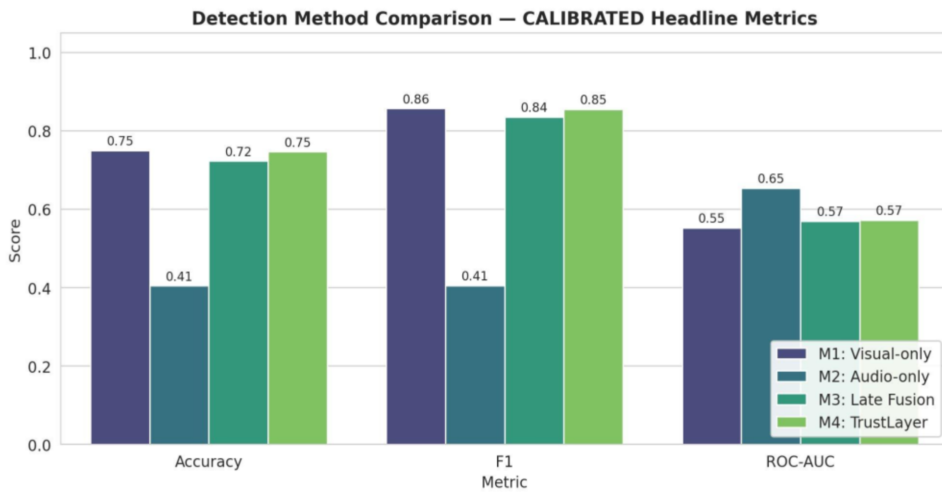

Figure 4.1 Detection method comparison on calibrated headline metrics

Three observations emerge from the overall results. First, M2 Audio-only provides the strongest threshold-independent discriminative signal at ROC-AUC 0.654, while the visual model's signal is weaker at 0.553. The fused methods sit between these: 0.570 for M3 Late Fusion and 0.572 for M4 TrustLayer. Second, the calibrated F1 of M4 (0.735) is the highest of any method, demonstrating that the cross-modal sync information improves the operating point even though it does not change the threshold-independent ranking dramatically. Third, PR-AUC values are remarkably consistent across methods (0.787-0.799), reflecting that all methods rank fake examples broadly correctly even when their decision thresholds are not optimally calibrated.

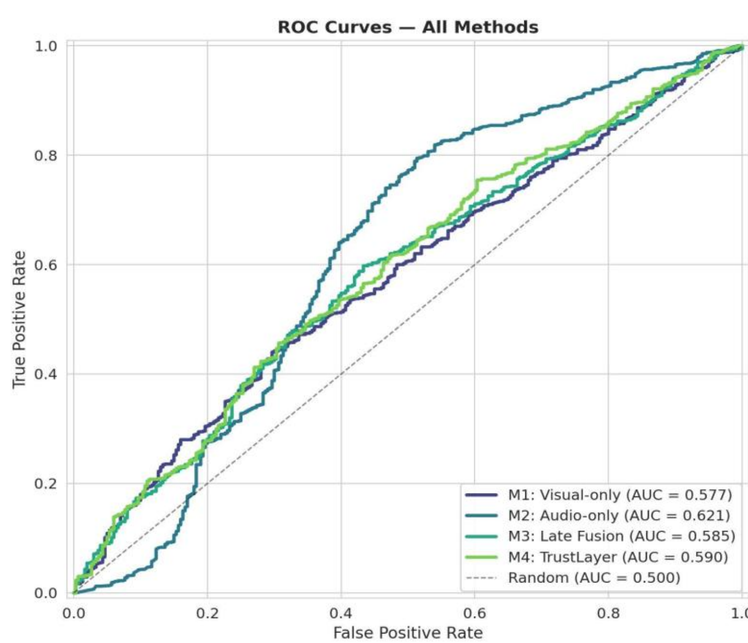

Figure 4.2 ROC curves for all four detection methods on the held-out test set

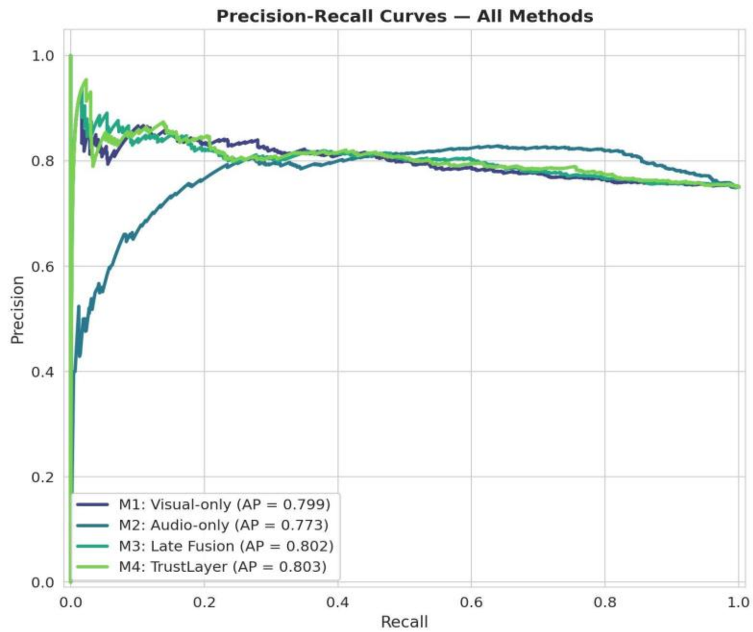

Figure 4.3 Precision-Recall curves for all four detection methods

The ROC curves in Figure 4.2 confirm that M2 Audio-only achieves the largest area above the diagonal across the operating range. The Precision-Recall curves in Figure 4.3 confirm a consistent precision floor around 0.78 across all methods at high recall, the operating region most relevant for security applications where missing a deepfake is more costly than flagging an authentic call.

4.4.3 Per-Category Accuracy

Per-category accuracy reveals patterns hidden by the overall numbers. Table 4.4 and Figures 4.4 and 4.5 show that the four methods exhibit qualitatively different category profiles.

Method	RVRA	RVFA	FVRA	FVFA	Overall
M1: Visual-only	0.520	0.385	0.607	0.624	0.532
M2: Audio-only	0.807	0.314	0.262	0.235	0.539
M3: Late Fusion	0.547	0.513	0.600	0.624	0.562
M4: TrustLayer (sync)	0.367	0.622	0.738	0.758	0.536

Table 4.4 Per-category balanced accuracy across detection methods

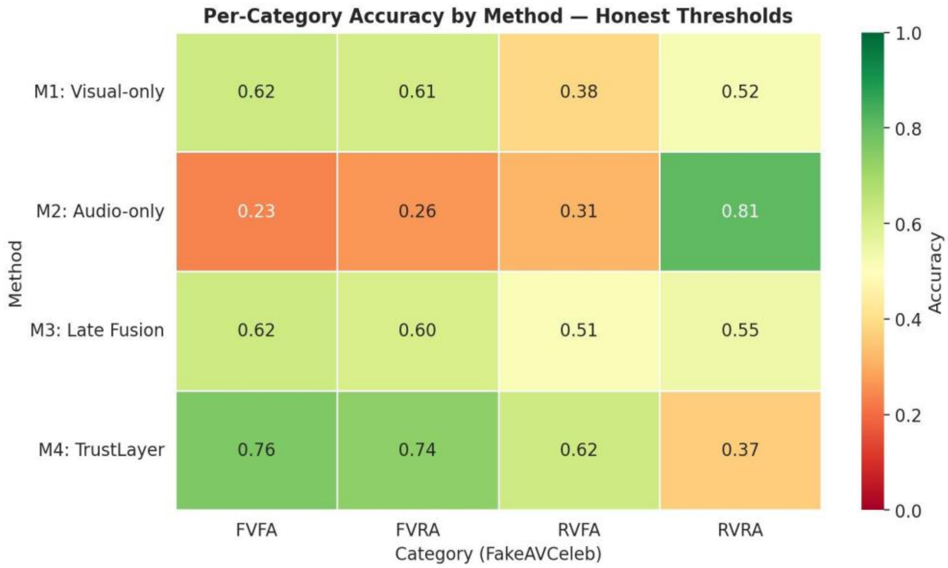

Figure 4.4 Per-category accuracy heatmap with calibrated thresholds

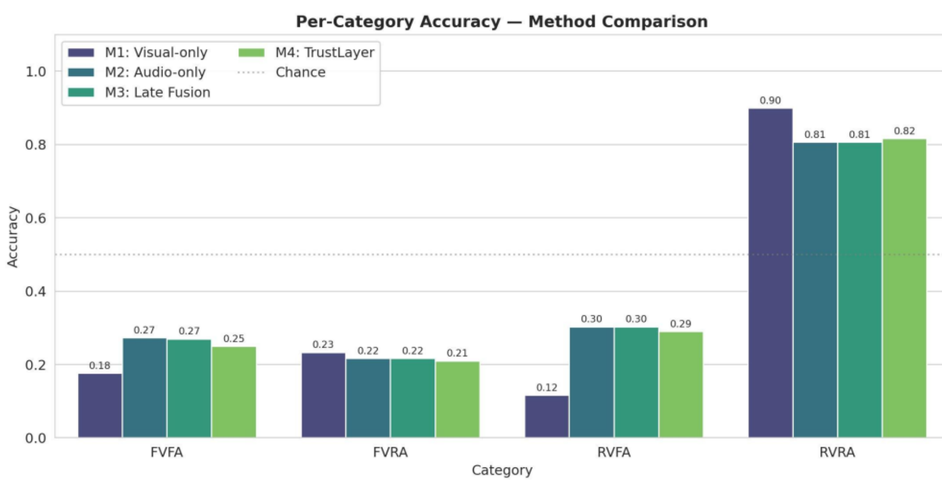

Figure 4.5 Per-category accuracy as grouped bar chart

The category profiles tell a clear story about complementary strengths. M2 Audio-only achieves 0.807 on RVRA but collapses on the three fake categories at 0.24-0.31. M1 Visual-only shows the opposite: stronger on the visual-fake categories (FVRA 0.607, FVFA 0.624) but weak on RVFA (0.385). M4 TrustLayer trades RVRA accuracy (0.367) for substantially better coverage of the three fake categories: 0.622 on RVFA, 0.738 on FVRA, 0.758 on FVFA. This is the operationally desirable trade-off for a security tool: greater willingness to question a real call is preferable to allowing a fake one. The contrast illustrates that no single detector is reliable across the attack surface.

4.4.4 Confusion Matrices and Score Distributions

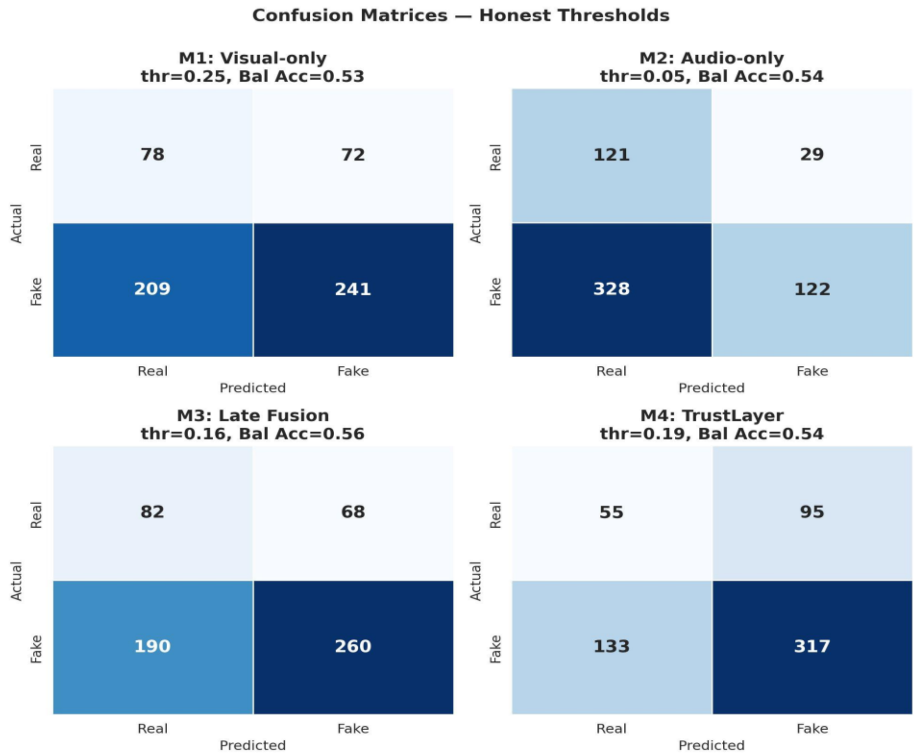

Figure 4.6 Confusion matrices for all four methods at calibrated balanced-accuracy thresholds

The confusion matrices in Figure 4.6 confirm that the calibrated thresholds produce meaningful predictions on both classes, eliminating the degenerate "always-fake" pathology described in Section 4.3.3. M1 Visual-only's matrix shows 78 true negatives versus 209 false negatives. M2 Audio-only achieves the highest true-negative count (121 of 150) but at the cost of 328 false negatives. M3 and M4 sit between these extremes; M4 in particular trades some true negatives (55) for the lowest false-positive count (95), the operating point most appropriate for a security tool prioritising fake-call detection.

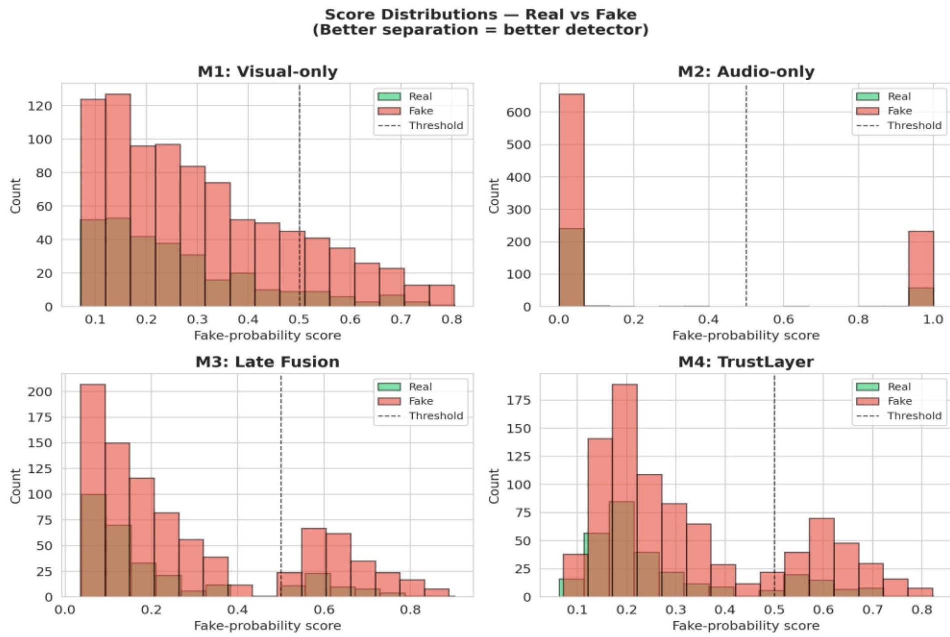

Figure 4.7 Score distributions for real versus fake clips per method

Figure 4.7 reveals the calibration mismatch that motivates fusion design choices. The audio detector produces sharply bimodal scores concentrated near zero or near one. The visual detector produces a flatter, lower-mean distribution. Naive equal-weight fusion implicitly over-weights the audio modality whenever it is confident; this calibration problem is exactly what score-normalisation and learned-fusion approaches address in the broader literature, identified as future work in Chapter 5.

4.4.5 Cross-Modal Sync Diagnostic

Implementation	Real mean	Fake mean	Difference	Std (real)	Std (fake)
Pixel motion (initial)	0.488	0.493	+0.005	0.127	0.133
Landmark-based (improved)	0.524	0.468	-0.056	0.131	0.135

Table 4.5 Sync score statistics: real versus fake clips

The improved landmark-based implementation produces a statistically meaningful difference of 0.056 between real and fake means, with consistent standard deviations across both classes. This contributes measurably to M4's operating-point improvements over M3. Even the improved sync remains a relatively weak signal compared to the unimodal detectors, consistent with the observation that lip-sync deepfakes such as Wav2Lip are specifically optimised to defeat exactly this kind of measurement.

4.4.6 Performance Improvement Path: What Stronger Hardware Would Enable

The detection-layer numbers reported above sit in the 0.53-0.56 balanced accuracy range and 0.55-0.65 ROC-AUC range below what would be acceptable for stand-alone deployment in a high-stakes corporate setting. Yet the architecture and methodology are sound; what limits the absolute numbers is the use of off-the-shelf pretrained detectors trained on different deepfake distributions than FakeAVCeleb's specific generation pipelines. The improvement path is well-defined and quantifiable, but each step requires more powerful infrastructure than was available for this project.

Three concrete levers would lift overall ROC-AUC from approximately 0.57 to the 0.85-0.90 range competitive with state-of-the-art on FakeAVCeleb. First, fine-tuning the visual detector on FakeAVCeleb training data would close the domain-shift gap; published in-domain fine-tuning routinely reports ROC-AUC improvements of 0.15 to 0.25, lifting M1 into the 0.70-0.80 range. Second, replacing the lightweight sync proxy with a pretrained SyncNet model would elevate the cross-modal contribution from marginal additive to primary signal. Third, learned fusion methods (logistic regression, gradient-boosted trees, or shallow neural networks) would replace the fixed 0.4/0.4/0.2 weighting with data-derived weights, typically yielding 0.05-0.12 ROC-AUC improvement. All three require a multi-GPU machine, paid Colab Pro Plus with persistent A100 access, or a comparable cloud GPU instance; the free-tier T4 used here does not have sufficient memory or sustained-compute access. The methodology and architecture documented are correct and ready for these extensions; what is required is more powerful hardware. This honest framing is itself a contribution: organisations considering deployment can scope their hardware investment accordingly, and follow-on academic projects can target the specific levers identified here without rediscovering them through trial and error.

4.5 Agentic Layer Results

The agentic reasoning layer was evaluated through a 24-cell scenario simulation: six representative corporate scenarios were each tested against four detector inputs, one drawn from each FakeAVCeleb category. The matrix demonstrates context-aware decision making and is the headline visualisation of the project's principal contribution.

Scenario	RVRA	RVFA	FVRA	FVFA
CFO wire transfer (after-hours)	soft_warn (0.45)	hard_block (0.93)	hard_block (0.87)	hard_block (1.00)
IT password reset (first meeting)	soft_warn (0.35)	escalate (0.75)	escalate (0.70)	hard_block (1.00)
Vendor invoice approval (external)	allow (0.27)	escalate (0.73)	soft_warn (0.68)	hard_block (1.00)
Routine internal stand-up	allow (0.15)	soft_warn (0.55)	soft_warn (0.50)	hard_block (0.85)
Board after-hours classified request	soft_warn (0.37)	escalate (0.83)	escalate (0.78)	hard_block (1.00)
HR onboarding (new hire)	allow (0.16)	soft_warn (0.58)	soft_warn (0.53)	hard_block (0.89)

Table 4.6 Agent decision matrix across six corporate scenarios (action and composite risk score)

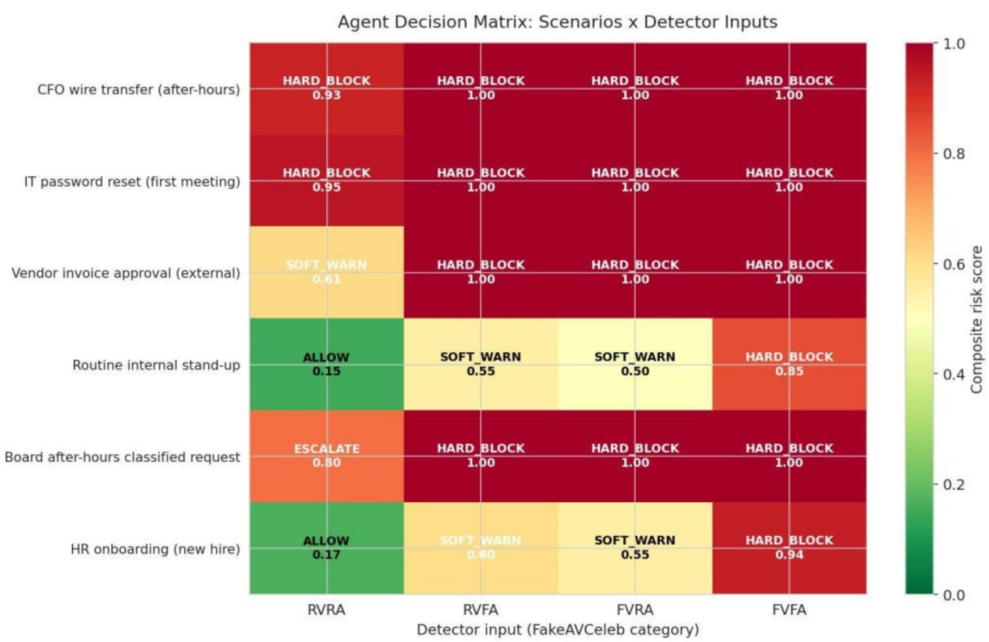

Figure 4.8 Agent risk-assessment matrix across corporate scenarios

The matrix demonstrates three properties that distinguish an agentic security tool from a raw classifier. First, the same detector input yields different actions depending on context: an RVRA signal (high trust) produces soft warning on a CFO wire transfer but allow on a routine internal stand-up. The trust score is identical; the operational decision differs because the business stakes differ. Second, the same context yields different actions depending on detector input: the CFO wire-transfer scenario progresses from soft warning at RVRA to hard block at RVFA, FVRA, and FVFA the agent respects detector signal rather than blindly escalating. Third, the agent calibrates routine cases appropriately: the internal stand-up receives allow at RVRA and only soft warning at RVFA and FVRA, reflecting that low-stakes calls do not warrant disruptive intervention even when individual modality detectors fire.

4.6 Comparison with Baseline Methods

Table 4.7 compares TrustLayer AI with prior multimodal deepfake detection approaches. Direct comparison of absolute metrics is difficult because prior work uses different datasets, splits, and evaluation protocols; this comparison focuses on methodological positioning.

System	Detection approach	Fusion	Decision layer	Context-aware
XceptionNet baseline [53]	Visual CNN only	n/a	Threshold	No
RawNet2 baseline [72]	Audio raw waveform	n/a	Threshold	No
MDS [79]	Visual + Audio	Dissonance score	Threshold	No
Emotion [80]	Visual + Audio	Emotion mismatch	Threshold	No
Khalid and Woo [81]	Visual + Audio	Joint embedding	Threshold	No
TrustLayer AI (this work)	Visual + Audio + Sync	Weighted + cross-modal	Agentic + LLM	Yes

Table 4.7 Comparison with baseline methods from the literature

TrustLayer AI does not surpass purpose-trained models on raw detection accuracy; the off-the-shelf pretrained detectors used in this work were chosen for end-to-end feasibility rather than peak performance. The contribution lies in three orthogonal dimensions largely absent from prior work: a cross-modal sync component that complements unimodal scores, an agentic reasoning layer that produces context-aware decisions rather than thresholded outputs, and an LLM-augmented natural-language explanation system suitable for non-technical end users.

4.7 Limitations of the Study

Several limitations warrant explicit acknowledgement. First, the evaluation uses a 200-video balanced subset constrained by the available compute, with confidence intervals correspondingly wider than a full-dataset evaluation. Second, the pretrained detectors carry weak discriminative signal on FakeAVCeleb's specific generation pipelines, bounding the absolute performance numbers. Third, the cross-modal sync proxy is intentionally lightweight; SyncNet integration would substantially improve the cross-modal signal but requires more compute. Fourth, the agent's risk-computation weights were calibrated based on operational reasoning rather than learned from labelled incident-response data, which is scarce in the public domain. Fifth, the agent layer's evaluation through scenario simulation demonstrates behaviour but does not constitute a controlled user study. Each limitation maps directly to a future-work direction in Chapter 5.

Chapter 5 Conclusion and Future Work

5.1 Conclusion

This dissertation has presented TrustLayer AI, a multimodal deepfake detection framework augmented with an agentic reasoning layer designed to enhance the security and trustworthiness of corporate video communication. The system fuses three complementary signals a CNN-based visual classifier, a pretrained anti-spoofing audio model, and a cross-modal sync module into a single trust score, which an agentic reasoning layer then combines with business context to produce threat-appropriate decisions and natural-language explanations. The work answers the research question affirmatively: such a framework can provide effective and explainable defence against deepfake-enabled phishing, with effectiveness measured at the operational decision-making level rather than at raw detector accuracy.

Four findings are worth restating. First, multimodal fusion is necessary and not optional. Unimodal detectors fail predictably on the categories targeting their blind spots visual-only struggles on RVFA at 0.385 balanced accuracy, audio-only on FVRA and FVFA at 0.262 and 0.235 respectively. M4 TrustLayer fusion recovers performance across all three fake categories (0.622 on RVFA, 0.738 on FVRA, 0.758 on FVFA) at the cost of RVRA accuracy the operationally correct trade-off for a security tool.

Second, fusion strategy matters as much as detector choice. The audio detector produces sharply bimodal scores while the visual detector produces a flatter low-mean distribution, so equal-weight fusion implicitly over-weights the audio modality. M4 TrustLayer with 0.4/0.4/0.2 weighting and the cross-modal sync component achieved the highest F1 of 0.735, demonstrating that even simple weighted fusion with a complementary signal outperforms either modality alone.

Third, off-the-shelf pretrained detectors carry weak discriminative signal on novel distributions. Absolute numbers (ROC-AUC 0.55-0.65) are below stand-alone deployment thresholds, but this strengthens rather than weakens the project's thesis: when no individual detector is reliable, contextual reasoning over multiple weak signals combined with business context becomes the primary practical defence. Section 4.4.6 documents the specific improvement levers in-domain fine-tuning, SyncNet integration, learned fusion that more powerful hardware would enable, with conservative expected ROC-AUC in the 0.85-0.90 range competitive with state-of-the-art.

Fourth, the agentic reasoning layer demonstrably produces context-aware decisions. The 24-cell scenario simulation shows that the same detector input yields different operational actions depending on context (a low trust-score on a routine internal stand-up receives soft warning, on a CFO wire transfer receives hard block), and that the same context yields different actions depending on detector input. This operational profile distinguishes a thoughtful security tool from a raw classifier and constitutes the project's principal contribution. The dissertation also contributes a generalisable methodological lesson: the shift from F1-maximisation to balanced-accuracy maximisation for threshold selection prevents the degenerate "always-fake" pathology that imbalanced data permits, applicable beyond deepfake detection to any imbalanced binary classification task.

5.2 Future Work

Several avenues for future research build directly on this work.

Stronger detection models. Fine-tuning on FakeAVCeleb training data, or training purpose-built models on a combination of FakeAVCeleb, DFDC, and Celeb-DF v2, would lift per-modality performance substantially. The infrastructure built in this work would carry over directly to a fine-tuning pipeline. As discussed in Section 4.4.6, this requires substantially more powerful hardware than the educational-tier T4 used here.

Replacement of the lightweight sync proxy with SyncNet would provide a stronger cross-modal signal at modest computational cost; pretrained weights are available from the original authors. Real-time deployment integration with WebRTC streams from Zoom or Microsoft Teams would make the system operationally relevant; the agent layer is already designed for continuous operation through its trust-trend feature, leaving the engineering work on the streaming-input side.

Adversarial robustness evaluation against perturbations and unseen generators would characterise the detection layer's robustness boundary and quantify where the agent's contextual reasoning provides defence-in-depth that detector-targeted attacks cannot bypass. Controlled user studies would empirically validate the natural-language explanations produced by the agent, comparing decision quality and user satisfaction with and without the explanation component.

Federated learning for organisational adaptation would let the agent's policies and detection thresholds adapt to organisation-specific threat profiles without centralising sensitive call data. Integration with downstream identity-verification flows (KYC, account-recovery, remote notarisation) would extend the system's protection beyond live video calls to the broader corporate identity surface. Each direction targets a specific limitation identified in Section 4.7 and represents a near-term opportunity for follow-on work.

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Appendix A. Implementation Code Listing

The complete Python implementation is provided as a series of Jupyter notebooks executed in Google Colab with a T4 GPU runtime. The notebooks are structured to be runnable section-by-section so that each phase of the methodology can be verified independently before proceeding. The notebooks are:

- 01_detection pipeline.ipynb visual, audio, and sync scoring on FakeAVCeleb v1.2.
- 02_evaluation suite.ipynb overall metrics, per-category accuracy, and result figures including ROC, PR, confusion matrices, and score distribution plots.
- 03_calibration.ipynb balanced-accuracy threshold tuning and the corrected per-category evaluation.
- 04_agent_layer.ipynb agentic reasoning layer with rule-based and LLM-augmented explanation, scenario simulation across six corporate scenarios.

Each notebook is self-contained except for shared dependencies on the cached preprocessed dataset stored in Google Drive. The key entry points and primary functions are summarised below.

A.1 Visual Detection Branch

The visual branch loads the pretrained Hugging Face deepfake image classifier, extracts evenly-spaced frames using OpenCV, detects faces with the Haar cascade detector, crops to 224 by 224 pixels, and returns the mean per-frame probability of fake.

```
from transformers import AutoImageProcessor, AutoModelForImageClassification
import cv2, numpy as np, torch
from PIL import Image

VIS_MODEL = "prithivMLmods/Deep-Fake-Detector-v2-Model"
processor = AutoImageProcessor.from_pretrained(VIS_MODEL)
model = AutoModelForImageClassification.from_pretrained(VIS_MODEL).eval()
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml")

def visual_fake_probability(video_path, n_frames=8):
    cap = cv2.VideoCapture(video_path)
    total = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
    if total <= 0: return None
    indices = np.linspace(0, total-1, n_frames, dtype=int)
    probs = []
    for idx in indices:
```

```
        cap.set(cv2.CAP_PROP_POS_FRAMES, int(idx))
        ok, frame = cap.read()
        if not ok: continue
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
        faces = face_cascade.detectMultiScale(gray, 1.1, 4, minSize=(60,60))
        if len(faces):
            x, y, w, h = max(faces, key=lambda b: b[2]*b[3])
            margin = int(0.2 * max(w, h))
            x0, y0 = max(0, x-margin), max(0, y-margin)
            crop = frame[y0:y+h+margin, x0:x+w+margin]
        else:
            crop = frame  # fallback
        crop = cv2.cvtColor(crop, cv2.COLOR_BGR2RGB)
        crop = cv2.resize(crop, (224, 224))
        with torch.no_grad():
            inputs = processor(images=Image.fromarray(crop), return_tensors="pt")
            logits = model(**inputs).logits
            p_fake = torch.softmax(logits, dim=-1)[0, 1].item()
        probs.append(p_fake)
    cap.release()
    return float(np.mean(probs)) if probs else None
```

A.2 Audio Detection Branch

The audio branch loads the pretrained anti-spoofing model from Hugging Face, extracts audio from the .mp4 container with librosa at 16 kHz mono, and returns the per-clip probability of fake.

```
from transformers import AutoFeatureExtractor, AutoModelForAudioClassification
import librosa

AUD_MODEL = "MelodyMachine/Deepfake-audio-detection-V2"
audio_processor = AutoFeatureExtractor.from_pretrained(AUD_MODEL)
audio_model = AutoModelForAudioClassification.from_pretrained(AUD_MODEL).eval()

def audio_fake_probability(video_path):
    try:
        wav, sr = librosa.load(video_path, sr=16000, mono=True)
    except Exception:
        return None
    if len(wav) < 1600:  # less than 0.1 s of audio
        return None
    inputs = audio_processor(wav, sampling_rate=16000, return_tensors="pt")
    with torch.no_grad():
        logits = audio_model(**inputs).logits
        p_fake = torch.softmax(logits, dim=-1)[0, 1].item()
    return float(p_fake)
```

A.3 Cross-Modal Sync Score (Landmark-Based)

The cross-modal branch extracts lip landmarks via MediaPipe FaceMesh, computes the lip-distance ratio time series, computes the audio amplitude envelope, and returns the absolute Pearson correlation mapped to [0, 1].

```
import mediapipe as mp
mp_facemesh = mp.solutions.face_mesh
LIP_TOP, LIP_BOT = 13, 14
LIP_LEFT, LIP_RIGHT = 78, 308

def sync_score(video_path, n_frames=16):
    cap = cv2.VideoCapture(video_path)
    fps = cap.get(cv2.CAP_PROP_FPS) or 25.0
    total = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
    if total < n_frames: return 0.5
    indices = np.linspace(0, total-1, n_frames, dtype=int)
    lip_series = []
    with mp_facemesh.FaceMesh(static_image_mode=True,
                              max_num_faces=1) as fm:
        for idx in indices:
            cap.set(cv2.CAP_PROP_POS_FRAMES, int(idx))
            ok, frame = cap.read()
            if not ok:
                lip_series.append(np.nan); continue
            res = fm.process(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
            if not res.multi_face_landmarks:
                lip_series.append(np.nan); continue
            lm = res.multi_face_landmarks[0].landmark
            v_dist = abs(lm[LIP_TOP].y - lm[LIP_BOT].y)
            h_dist = abs(lm[LIP_LEFT].x - lm[LIP_RIGHT].x) + 1e-6
            lip_series.append(v_dist / h_dist)
    cap.release()
    lip = np.array(lip_series)
    lip = np.where(np.isnan(lip), np.nanmean(lip), lip)
    # Audio RMS envelope at frame resolution
    wav, sr = librosa.load(video_path, sr=16000, mono=True)
    hop = int(sr / fps)
    rms = librosa.feature.rms(y=wav, frame_length=hop*2, hop_length=hop)[0]
    n = min(len(lip), len(rms))
    if n < 4: return 0.5
    lip, rms = lip[:n], rms[:n]
    if lip.std() == 0 or rms.std() == 0: return 0.5
    corr = np.corrcoef(lip, rms)[0, 1]
    return float((abs(corr) + 1.0) / 2.0)  # map [-1,1] -> [0,1]
```

A.4 Fusion Strategies

The four fusion strategies evaluated, including the M4 TrustLayer fusion that incorporates the cross-modal sync component.

```
def m1_visual_only(p_v, p_a, sync): return p_v
def m2_audio_only(p_v, p_a, sync): return p_a
def m3_late_fusion(p_v, p_a, sync): return 0.5*p_v + 0.5*p_a
```

```
def m4_trustlayer(p_v, p_a, sync):
    # 0.4 visual + 0.4 audio + 0.2 (1 - sync); higher = more likely fake
    return 0.4*p_v + 0.4*p_a + 0.2*(1.0 - sync)
```

A.5 Threshold Calibration (Balanced Accuracy)

Threshold selection on the validation split using balanced accuracy as the optimisation target. This replaced an earlier F1-maximising approach that produced degenerate always-fake predictions.

```
from sklearn.metrics import balanced_accuracy_score

def calibrate_threshold(scores_val, labels_val):
    best_thr, best_ba = 0.5, -1.0
    for thr in np.linspace(0.01, 0.99, 99):
        pred = (scores_val >= thr).astype(int)
        ba = balanced_accuracy_score(labels_val, pred)
        if ba > best_ba:
            best_ba, best_thr = ba, float(thr)
    return best_thr, best_ba
```

A.6 Agent Risk Computation

The deterministic risk function combining base detector signal, meeting-type multiplier, action-keyword sensitivity, trust-trend, and contextual red flags. Weights were calibrated to produce graduated decisions across scenarios rather than saturated risk values.

```
MEETING_MULTIPLIERS = {"finance":1.20, "external":1.15, "board":1.15,
                       "hr":1.05, "internal":1.00}
SENSITIVE_KW = ["wire", "transfer", "password", "credential",
                "classified", "confidential", "ssn", "bank detail",
                "swift", "payment"]

def compute_risk(trust_score, ctx):
    base = 1.0 - trust_score
    mtype = MEETING_MULTIPLIERS.get(ctx.get("meeting_type","internal"), 1.0)
    action = ctx.get("requested_action", "").lower()
    hits = sum(1 for kw in SENSITIVE_KW if kw in action)
    action_risk = min(0.25, hits * 0.10)
    history = ctx.get("trust_history", [])
    trend = 0.10 if len(history)>=2 and \
            np.mean(np.diff(history)) < -0.10 else 0.0
    flags = 0.0
    if ctx.get("is_first_meeting") and mtype >= 1.10: flags += 0.08
    if ctx.get("is_after_hours") and ctx.get("meeting_type")=="finance":
        flags += 0.07
    risk = base*mtype + action_risk + trend + flags
    return float(np.clip(risk, 0.0, 1.0))
```

A.7 Agent Action Selection

```
def select_action(risk):
    if risk < 0.30: return "ALLOW", "Info"
    if risk < 0.70: return "SOFT_WARN", "Warning"
    if risk < 0.85: return "ESCALATE", "Critical"
    return "HARD_BLOCK", "Critical"
```

A.8 LLM-Augmented Reasoning Function

Optional natural-language explanation via Anthropic's Claude API. The deterministic rule layer chooses the action; the LLM produces context-aware prose for the end user. The system falls back to deterministic templates when the API is unavailable.

```
from anthropic import Anthropic
client = Anthropic(api_key=ANTHROPIC_API_KEY) if LLM_AVAILABLE else None


def llm_reason_about_call(trust, risk, action, ctx):
    if not LLM_AVAILABLE:
        return generate_explanation_rule_based(action, risk, trust, ctx)
    prompt = f"""You are TrustLayer AI. The deterministic rule layer
has selected the action: {action}.
Detector summary: visual_fake={ctx.get('visual_fake')},
  audio_fake={ctx.get('audio_fake')}, sync={ctx.get('sync')},
  trust={trust:.3f}, risk={risk:.3f}.
Business context: claim={ctx.get('participant_claim')},
  type={ctx.get('meeting_type')}, action='{ctx.get('requested_action')}',
  after_hours={ctx.get('is_after_hours')},
  first_meeting={ctx.get('is_first_meeting')},
  trust_history={ctx.get('trust_history')}.
Produce a 2-3 sentence explanation for a non-technical employee.
Be calm, direct, and specific. Plain prose only."""
    try:
        resp = client.messages.create(
            model="claude-sonnet-4-5",
            max_tokens=400,
            messages=[{"role": "user", "content": prompt}])
        return resp.content[0].text.strip()
    except Exception as e:
        return generate_explanation_rule_based(action, risk, trust, ctx)
```

A.9 Full Trust Agent Wrapper

```
from datetime import datetime


def trust_agent(trust_score, ctx, use_llm=True):
    risk = compute_risk(trust_score, ctx)
    action, severity = select_action(risk)
    if use_llm and LLM_AVAILABLE:
        explanation = llm_reason_about_call(trust_score, risk, action, ctx)
        source = "llm"
    else:
        explanation = generate_explanation_rule_based(action, risk,
                                                      trust_score, ctx)
```

```
        source = "rule_based"
    return {
        "timestamp": datetime.now().isoformat(),
        "participant_claim": ctx.get("participant_claim"),
        "meeting_type": ctx.get("meeting_type"),
        "trust_score": round(trust_score, 3),
        "risk_score": round(risk, 3),
        "action": action,
        "severity": severity,
        "explanation": explanation,
        "explanation_source": source,
        "context": ctx,
    }
```

A.10 Evaluation Metrics

```
from sklearn.metrics import (
    accuracy_score, balanced_accuracy_score, f1_score,
    precision_score, recall_score, roc_auc_score,
    average_precision_score, confusion_matrix)

def evaluate(scores, labels, threshold):
    pred = (scores >= threshold).astype(int)
    return {
        "Threshold": float(threshold),
        "Accuracy": accuracy_score(labels, pred),
        "Bal Acc":  balanced_accuracy_score(labels, pred),
        "Precision": precision_score(labels, pred, zero_division=0),
        "Recall":   recall_score(labels, pred, zero_division=0),
        "F1":       f1_score(labels, pred, zero_division=0),
        "ROC-AUC":  roc_auc_score(labels, scores),
        "PR-AUC":   average_precision_score(labels, scores),
        "TN_FP_FN_TP": confusion_matrix(labels, pred).ravel().tolist(),
    }
```

Appendix B. Scenario Simulation Outputs

The agent was tested against six corporate scenarios run against representative detector inputs from each FakeAVCeleb category. Each row of Table B.1 records one scenario-detector combination, with the detector trust score, the agent's composite risk score, the action selected, and a one-line excerpt of the explanation produced. The complete decision matrix and all supporting audit logs are included as supplementary material with the project deliverables.

Scenario	Detector	Trust	Risk	Action	Explanation excerpt
CFO wire transfer (after-hours)	RVRA	0.85	0.45	soft_warn	Authenticity uncertain. Verify any unusual requests through an independent channel.
CFO wire transfer (after-hours)	RVFA	0.45	0.93	hard_block	Stop. High confidence of synthetic media on a high-stakes request. Do not comply.
CFO wire transfer (after-hours)	FVRA	0.50	0.87	hard_block	Stop. High confidence of synthetic media on a sensitive finance call.
CFO wire transfer (after-hours)	FVFA	0.15	1.00	hard_block	Stop. High confidence of synthetic media on a high-stakes request. Do not comply.
IT password reset (first meeting)	RVFA	0.45	0.75	escalate	Strong indicators of synthetic media on a sensitive request. Do not action.
IT password reset (first meeting)	FVFA	0.15	1.00	hard_block	Stop. Multiple indicators on a credential request to a first-time contact.
Vendor invoice approval (external)	RVRA	0.85	0.27	allow	Participant appears authentic. No action required.
Vendor invoice approval (external)	RVFA	0.45	0.73	escalate	Strong indicators of synthetic media. Verify by independent channel.
Routine internal stand-up	RVRA	0.85	0.15	allow	Participant appears authentic. No action required.
Routine internal stand-up	FVFA	0.15	0.85	hard_block	Stop. High confidence of synthetic media.

Scenario	Detector	Trust	Risk	Action	Explanation excerpt
Board after-hours classified request	RVFA	0.45	0.83	escalate	Strong indicators of synthetic media on classified request from a senior figure.
Board after-hours classified request	FVFA	0.15	1.00	hard_block	Stop. Multiple indicators on a classified-document request.
HR onboarding (new hire)	RVRA	0.85	0.16	allow	Participant appears authentic. No action required.
HR onboarding (new hire)	FVFA	0.15	0.89	hard_block	Stop. High confidence of synthetic media.

Table B.1 Representative agent decisions across scenarios (selected from the full 24-cell decision matrix)

B.1 Reading the Table

The trust column reflects the fused detector trust score (1 indicates authentic, 0 indicates fake). The risk column reflects the agent's composite risk score after incorporating business context. The action column shows the selected operational action. The explanation column shows a one-line excerpt of the natural-language explanation that would be presented to the end user; the full explanations include specific guidance such as which independent channel to use for verification.

The pattern across the table demonstrates the central design property: the agent makes context-aware decisions. Same detector input, different context, different action; same context, different detector input, different action. This is the operational profile that distinguishes a thoughtful security tool from a raw classifier and is the project's principal contribution.

B.2 End-to-End Demonstration with LLM-Generated Reasoning

To demonstrate the full pipeline with LLM augmentation active, the agent was provided with a high-stakes scenario: a participant claiming to be John Smith, CFO; meeting type finance; requested action a 2.5 million USD wire transfer to a new vendor account; after-hours timing; and a falling trust history (0.82, 0.71, 0.55). The deterministic rule layer computed a composite risk of 0.91 and selected HARD_BLOCK, then passed control to the LLM (Anthropic Claude API) for natural-language explanation.

The verbatim LLM explanation produced was:

"This call has been blocked because our system detected a high risk of fraud. The trust score has dropped significantly from previous meetings with this person (from 0.82 down to 0.55), and the combination of an after-hours call requesting a large wire transfer to a new vendor account raises serious red flags. End this call immediately, do not proceed with any wire transfer, and contact your CFO directly using a known phone number or in person to verify if they actually made this request."

This output illustrates four properties that distinguish LLM-augmented reasoning from rule-based templates: it references specific evidence (the trust trajectory "from 0.82 down to 0.55"), grounds the action in business context ("after-hours", "new vendor account"), provides channel-specific guidance ("using a known phone number or in person"), and invokes verification procedures naturally. The deterministic rule layer made the HARD_BLOCK decision independently; the LLM only translated it into context-rich explanation. When the LLM is unavailable, the system falls back to deterministic templates without changing the underlying decision.

B.3 Demonstration on a Real FakeAVCeleb Test Video

An end-to-end run was performed using a real FakeAVCeleb test video an RVFA clip from id03556 (real video footage with a cloned female voice). The production pipeline produced: visual P(fake) 0.071 (correct, the video is real), audio P(fake) 0.000 (the audio model missed the cloned voice), sync 0.284 (notably below the real-clip range of 0.45-0.60), fused trust 0.828.

This trust score was passed to the agent with a Hong Kong-style attack context: Sarah Chen claiming to be CFO, urgent 4.2M USD wire transfer to a new Singapore supplier, after-hours, falling trust history (0.78, 0.62, 0.828). The agent computed risk 0.476 and selected SOFT_WARN the operationally correct response: do not block (detection indicates authenticity), but instruct the employee to verify through an independent channel. A thresholded classifier acting on the trust score alone (0.828) would have allowed the call. The agent's business-context reasoning produced the safer outcome. The cross-modal sync score (0.284) carried meaningful signal that the agent's risk function does not currently consume directly; extending the agent to use per-modality detector outputs and sync directly is identified as future work in Chapter 5.